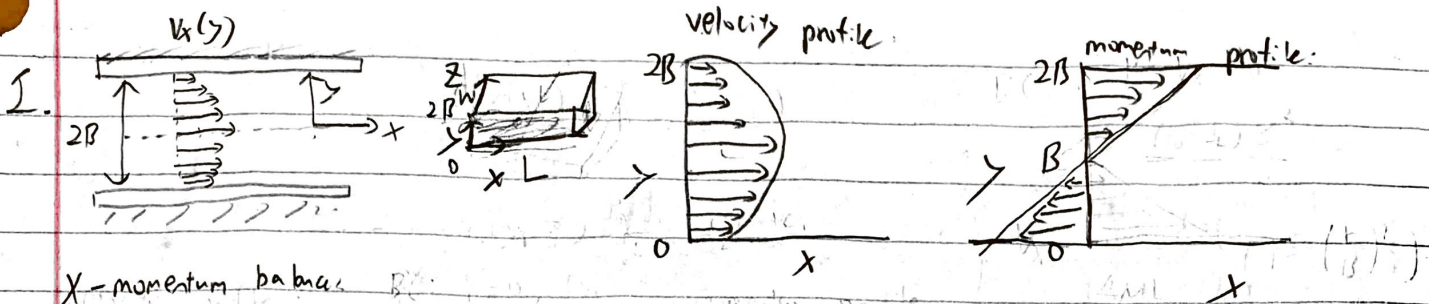




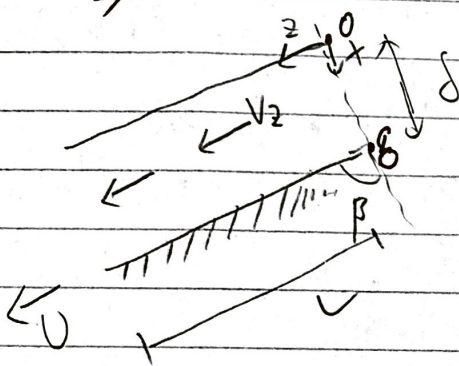
x-momentum balance

$$\phi_{yx}|_{y=0}(LW) - \phi_{yx}|_{y=2B}(LW) + \phi_{zx}|_{z=0}(2BL) - \phi_{zx}|_{z=L}(2BL) + \phi_{xx}|_{x=0}(W2B) - \phi_{xx}|_{x=L}(W2B) = 0$$

$$= T_{yx}|_{y=0}(LW) - T_{yx}|_{y=2B}(LW) + (P + \rho v_x v_x)2WB - (P + \rho v_x v_x)2WB = 0$$

$$\Rightarrow \frac{dT_{yx}}{dy} = \frac{P_0 - P_L}{L} \Rightarrow T_{yx} = \frac{P_0 - P_L}{L} y + C_1 \quad \Rightarrow v_{yx} = -\frac{P_0 - P_L}{4\mu} \left(\frac{y^2}{2} - By \right) = U$$

2.



$$v_{yx} = -\frac{P_0 - P_L}{4\mu} \left(\frac{y^2}{2} - By \right) \Rightarrow T_{yx} = \frac{P_0 - P_L}{L} (y - B)$$

BC: $y=0, v_x=0, y=2B, v_x=0$

$C_2 = 0, C_1 = -\frac{P_0 - P_L}{L} B$

$$v_2 = \frac{\rho g \cos \theta \delta^3}{2\mu} \left(1 - \left(\frac{x}{\delta} \right)^2 \right) + U$$

$$v_2 = \frac{\rho g \cos \theta \delta^3}{2\mu} + U$$

$$V_2 = U \text{ at } x = \delta \text{ (no slip)}$$

$$T_{xz} = 0 \text{ at } x = 0 \text{ (gas and fluid interface)}$$

$$\frac{dT_{xz}}{dx} = \rho g \cos \theta$$

$$T_{xz} = \rho g \cos \theta x + C_1$$

$$\text{BC: } T_{xz} = 0 \text{ at } x = 0$$

$$C_1 = 0$$

$$T_{xz} = \rho g \cos \theta x = -\mu \frac{dv_2}{dx}$$

$$\frac{dv_2}{dx} = -\frac{\rho g \cos \theta x}{\mu}$$

$$V_2 = -\frac{\rho g \cos \theta x^2}{2\mu} + C_2$$

$$\text{BC: } V_2 = U \text{ at } x = \delta$$

$$C_2 = U + \frac{\rho g \cos \theta \delta^2}{2\mu}$$

3 $\omega = \Omega$, $T = T_z$ $\theta = \psi_0 \approx 1^\circ$

$V = \omega r = \Omega r$

$b = r \sin \psi_0 \approx r \psi_0$

ϕ - direction momentum:

a) ϕ - direction momentum: $\phi(r, \theta) = \phi(r, \theta) - \phi(r, \theta) = \phi(r, \theta) - \phi(r, \theta) = \phi(r, \theta) - \phi(r, \theta)$

$\Rightarrow T_{\theta\phi} = -\frac{1}{r} \frac{\partial}{\partial \theta} (V_{\phi}) = C_1$

$\Rightarrow V_{\phi} = -\frac{r C_1}{\Omega} \theta + C_2$

BC: $V_{\phi} = \Omega r$, $\theta = \frac{\pi}{2} - \psi_0$

$V_{\phi} = 0$, $\theta = \frac{\pi}{2}$

$C_1 = \frac{\Omega \mu}{\psi_0}$, $C_2 = \frac{\Omega r \mu}{2 \psi_0}$

$\Rightarrow V_{\phi} = \Omega r \left(\frac{\pi}{2} - \theta \right)$

$\Rightarrow \frac{V_{\phi}}{r} = \Omega \left(\frac{\pi}{2} - \theta \right)$

$V_{\phi} = -\frac{1}{\Omega} \theta \left(\frac{\Omega \mu}{\psi_0} \right) + \frac{\Omega \mu}{2 \psi_0}$
 $= \frac{\Omega \mu}{2 \psi_0} - \frac{\Omega \mu}{\psi_0} \theta$
 $= \Omega r \left(\frac{\pi}{2} - \theta \right)$

b) $T_{\theta\phi} = -\frac{1}{r} \left[\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{V_{\phi}}{\sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \right]$

$= -\frac{1}{r} \left[\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\frac{V_{\phi}}{\sin \theta} \right) \right]$

$= -\frac{1}{r} \frac{1}{r} \frac{\partial}{\partial \theta} \left(\Omega r \left(\frac{\pi}{2} - \theta \right) \right) = -\frac{1}{r} \cdot \Omega r \cdot -\frac{1}{\psi_0}$

$= \frac{\Omega \mu}{\psi_0}$

c) $T_z = F \cdot L = T_{\theta\phi} \cdot A \cdot r = \int_0^{2\pi} \int_0^R (T_{\theta\phi} \cdot r) dr d\phi = \int_0^{2\pi} \int_0^R \left(\frac{\Omega \mu}{\psi_0} \right) r^2 dr d\phi$

$= \int_0^{2\pi} \frac{\Omega \mu}{\psi_0} \frac{r^3}{3} \Big|_0^R d\phi$

$= \int_0^{2\pi} \frac{\Omega \mu}{\psi_0} \frac{R^3}{3} d\phi$

$= \frac{2\pi}{3} \frac{\Omega \mu}{\psi_0} R^3$

$\mu = \frac{3 T_z \psi_0}{2\pi \Omega R^3}$

d) $R = 0.1 \text{ m}$, $\psi_0 = 0.5^\circ$, $\Omega = 10 \text{ rad/min}$, $\mu = 100 \text{ mPa}$

$\Omega = \frac{10 \text{ rad}}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ s}} = 0.167 \text{ rad/s}$

$T_z = \frac{2\pi}{3} \frac{\Omega \mu}{\psi_0} R^3$

$= \frac{2\pi}{3} \frac{100 \text{ mPa} \cdot 0.167 \text{ rad/s}}{0.5^\circ \times \frac{\pi \text{ rad}}{180^\circ}} (0.1 \text{ m})^3$

$= 4 \times 10^{-3} \text{ Nm} \times \frac{100 \text{ cm}}{1 \text{ m}} \times \frac{10000 \text{ dyn}}{1 \text{ N}}$

$= 40000 \text{ dyn cm}$